

Credit - Drishti Ias

## Montreal Protocol and World Ozone Day



Source: TH

### Why in News?

On the occasion of the **International Day for the Preservation of the Ozone Layer (World Ozone Day , observed on 16 th September)** , global attention was drawn to the historic journey of the **Montreal Protocol**.

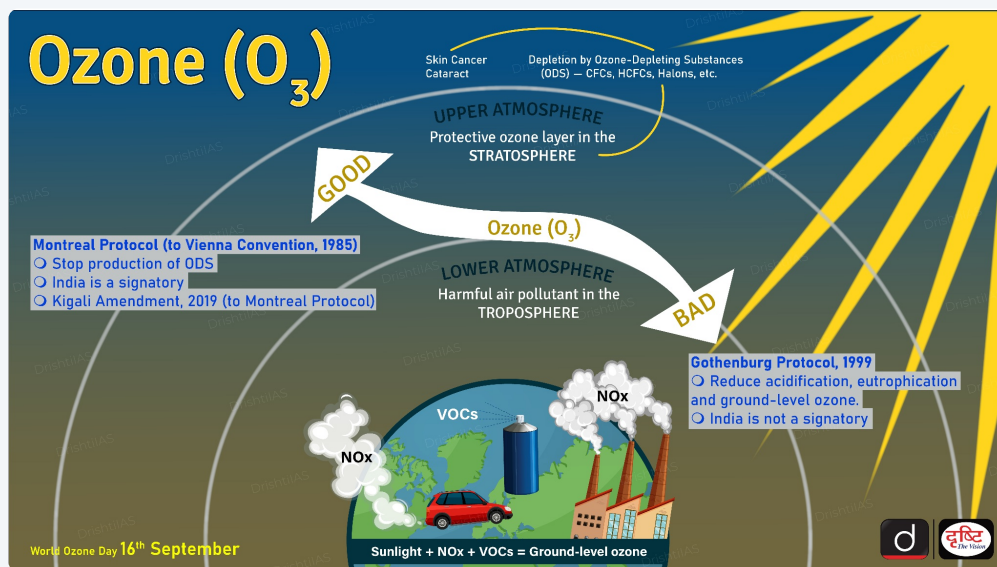
### What is the Montreal Protocol?

- **Background:** In the 1970s and 1980s, scientists confirmed that certain **synthetic chemicals**, primarily **chlorofluorocarbons (CFCs)** used in refrigeration and aerosols, were migrating to the **stratosphere** (altitude range of 12 to 50 kilometers above Earth's surface) and destroying **ozone molecules**, creating a massive "**ozone hole**" over **Antarctica**.
- **Vienna Convention (1985):** Prompted by this scientific discovery, the international community adopted the **Vienna Convention for the Protection of the Ozone Layer**.
  - Established a framework for international cooperation, environmental monitoring and scientific research, paving the way for the Montreal Protocol.
  - India became a Party to the Vienna Convention for the Protection of the Ozone Layer in 1991.
- **Montreal Protocol:** Adopted in 1987, it is a **legally binding multilateral environmental agreement** aimed at protecting the ozone layer by regulating the production and consumption of **nearly 100 man-made chemicals known as Ozone Depleting Substances (ODS)**, such as chlorofluorocarbons (CFCs) and **hydrochlorofluorocarbons (HCFCs)**.
  - **India became a Party to the Montreal Protocol in June 1992.**
  - **Regulatory Metric:** Under the Protocol, "Consumption" is formally defined as **Production + Imports - Exports - Verified Destruction**.
  - **Targeted Phase-Out:** The treaty categorizes chemicals by family and sets strict, differentiated timetables for developed and developing countries to eventually eliminate their use.
  - **Success:** Often celebrated as the world's most successful environmental treaty, it has facilitated the phase-out of 99% of ODS globally.
    - India has **met its Montreal Protocol commitments** , including the **complete phase-out of HCFC-141b** from production, import and use from **1 January 2020** . It also phased out **Chlorofluorocarbons (CFCs)**, **Carbon tetrachloride (CTC)** and **Halons** , achieving a **67.5% reduction in HCFC production and consumption by 2025** .

- **Universal Ratification:** On 16 th September 2009, the **Vienna Convention and the Montreal Protocol** achieved a monumental milestone by **becoming the first treaties in UN history to attain universal ratification (all 198 UN member states)**.
- **The Kigali Amendment (2016):** As industries moved away from ozone-damaging CFCs and HCFCs, they adopted **Hydrofluorocarbons (HFCs)** as alternatives.
  - While HFCs do not harm the ozone layer, they are exceptionally potent **Greenhouse Gases (GHGs)** that heavily exacerbate climate change.
  - To address this, the Parties to the Montreal Protocol adopted the **Kigali Amendment** at their 28 th Meeting of the Parties (MOP 28) in Kigali, Rwanda, in 2016.
  - It legally expanded the Protocol's mandate to **phase down non-ozone-depleting HFCs** (potent greenhouse gases with high global warming potential) turning the ozone agreement into an indispensable climate change mitigation framework by formally mandating a global phase-down of HFCs.
- **India's Commitment :** India ratified the **Kigali Amendment in 2021** , committing to **freeze HFC consumption in 2028** and phase it down by **85% by 2047** , in four stages: **10% by 2032, 20% by 2037, 30% by 2042 and 85% by 2047** against the baseline.
- **Climate Impact:** By curbing HFC emissions and pushing industries toward sustainable cooling alternatives, the successful implementation of the Kigali Amendment is projected to avoid up to **0.3-0.5°C of future global warming** by the end of this century.

# What is World Ozone Day?

- **Background:** In 1994, the **United Nations General Assembly (UNGA)** officially proclaimed 16<sup>th</sup> September as the International Day for the Preservation of the Ozone Layer.
  - The date was specifically chosen to commemorate the signing of the landmark **Montreal Protocol on Substances that Deplete the Ozone Layer in 1987**.
- **Objective:** The day serves as a powerful reminder of the ozone layer's vital role in shielding life on **Earth from the Sun's harmful ultraviolet (UV) radiation** and highlights the ongoing need to prevent environmental degradation.
- **2026 Theme & Focus:**
  - **Theme:** "Global action for a cooler planet - Celebrating 10 years of sustainable cooling under the Kigali Amendment to the Montreal Protocol" .
  - **Focus:** The theme highlights rising global demand for cooling amid increasing temperatures and calls for energy-efficient, climate-friendly cooling solutions, while marking a decade of efforts to phase down potent greenhouse gases.
- **India's Net Zero Portal and NAPCC Dashboard:** Launched as digital platforms to track and showcase India's climate commitments.
  - **Net Zero Portal:** Enables industries, organisations and government bodies to voluntarily register and report progress towards net-zero emissions.
  - **NAPCC Dashboard:** Provides a centralised gateway to track progress and insights across India's National Missions on Climate Change.



# Frequently Asked Questions (FAQs)

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## 1. What is the Montreal Protocol?

The **Montreal Protocol, adopted in 1987**, is a legally binding environmental treaty that regulates and phases out **ozone-depleting substances (ODS)**.

## 2. What is the significance of the Vienna Convention?

The **Vienna Convention (1985)** established the framework for **international cooperation, scientific research and monitoring** for ozone protection, paving the way for the Montreal Protocol.

## 3. What is the Kigali Amendment?

Adopted in **2016**, it mandates the **phasedown of HFCs**, potent greenhouse gases, thereby linking **ozone protection with climate change mitigation**.

## 4. When did India join the Montreal Protocol and ratify the Kigali Amendment?

India became a Party to the **Montreal Protocol in 1992** and ratified the **Kigali Amendment in 2021**.

## 5. What is the significance of World Ozone Day?

Observed on **16 September**, it raises awareness about **ozone-layer protection, UV radiation and sustainable cooling**, commemorating the Montreal Protocol.

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/6byTU3FMjpk>

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## UPSC Civil Services Examination Previous Year Question (PYQ)

**Q. Which one of the following is associated with the issue of control and phasing out of the use of ozone depleting substances? (2015)**

- (a) Bretton Woods Conference
- (b) Montreal Protocol
- (c) Kyoto Protocol
- (d) Nagoya Protocol

**Ans: (b)**

**Q. Consider the following statements: (2012)**

**Chlorofluorocarbons, known as ozone-depleting substances, are used**

1. in the production of plastic foams
2. in the production of tubeless tyres
3. in cleaning certain electronic components
4. as pressurizing agents in aerosol cans

**Which of the statements given above is/are correct?**

- (a) 1, 2 and 3 only
- (b) 4 only
- (c) 1, 3 and 4 only
- (d) 1, 2, 3 and 4

**Ans: (c)**

**Q. Consider the following: (2019)**

1. Carbon monoxide
2. Methane
3. Ozone
4. Sulphur dioxide

**Which of the above are released into the atmosphere due to the burning of crop/biomass residue?**

- (a) 1 and 2 only
- (b) 2, 3 and 4 only
- (c) 1 and 4 only
- (d) 1, 2, 3 and 4

**Ans: (d)**

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**Q. The formation of ozone holes in the Antarctic Region has been a cause of concern. What could be the reason for the formation of this hole? (2011)**

- (a) Presence of prominent tropospheric turbulence; and inflow of chlorofluorocarbons
- (b) Presence of prominent polar front and stratospheric clouds; and inflow of chlorofluorocarbons
- (c) Absence of polar front and stratospheric clouds; and inflow of methane and chlorofluorocarbons
- (d) Increased temperature at polar region due to global warming

**Ans: B**

# Pygmy Hog



Source: TH

The **Assam government recently highlighted the success of the Pygmy Hog Conservation Programme** , which has increased the pygmy hog population in the State significantly over the last three decades.

## Pygmy Hog

- **About:** The **pygmy hog ( *Porcula salvinia* )** is the **world's smallest wild suid** and the only surviving species of the genus **Porcula** . It is an **endemic species** of the Indian subcontinent and is closely associated with tall, wet alluvial grasslands of the **Himalayan foothills**.
- **Distribution:** Historically, it was found from **Uttar Pradesh to Assam** , including **Nepal's Terai and Bengal's Duars** . After remaining unreported for decades and being feared extinct, the **pygmy hog was rediscovered in 1971 in Assam** , with its surviving wild populations now mainly restricted to Assam, especially the Manas landscape.
- **Physical Features:** It is the smallest member of the **pig family (Suidae)**, characterised by an **extremely small body, ears and tail, short medial false hooves**, a streamlined body shape, and absence of facial warts found in many other pigs.
- **Habitat:** The pygmy hog **inhabits undisturbed tall grasslands of riverine ecosystems, comprising dense grasses, herbs, shrubs and young trees** .
  - Its fertile alluvial grassland habitat is **highly vulnerable to agricultural expansion and human settlement**, leading to habitat loss. It serves as an indicator species of healthy tall grassland ecosystems.
- **Diet:** It is an **omnivorous species** feeding on **roots, tubers, shoots, ground vegetation, worms, invertebrates and occasionally small vertebrates such as reptiles and eggs/nestlings of ground-nesting birds** .
- **Social Organisation:** Adult males are generally **solitary** , while females and juveniles live in **small family groups of about 4-6 individuals** ; males associate with females mainly during the breeding season.
- **Reproduction:** Breeding is **highly seasonal** , with births peaking during **late April-May with the onset of monsoon in Assam** . Litter size ranges from **2-6 piglets (usually 3-4)** , and both sexes construct nests throughout the year.
- **Conservation Status:**
  - **IUCN Red List : Endangered**
  - **CITES : Appendix I**

- [Wildlife \(Protection\) Act, 1972](#) : Schedule I
- **Conservation Efforts:** The **Pygmy Hog Conservation Programme (PHCP)** was initiated in **1995** , with major captive breeding efforts beginning in **1996** . The programme focuses on **captive breeding, disease prevention and reintroduction** .
- **Significance:** Conservation of pygmy hog habitat supports other threatened grassland species such as **Bengal florican, hispid hare, hog deer and greater one-horned rhinoceros** .



Read More: [Pygmy Hog](#)

## Anant Nag Selected for Dadasaheb Phalke Award 2024



**Source: PIB**

Veteran Kannada actor **Anant Nag** has been selected for the prestigious **Dadasaheb Phalke Award** , India's highest honour in cinema, making him the **second Kannada actor** to receive the award after **Dr. Rajkumar (1995)** .

- He will be presented with the award at the **72nd National Film Awards Ceremony** on **22nd September 2026 at Kevadia, Gujarat** .

## Anant Nag

- **Birth:** Born as **Anant Nagarkatte in 1948 in Karnataka** , he has acted in **300+ films**, including **over**

**250 Kannada films** , across mainstream and parallel cinema.

- **Major Works:** He made his Kannada film debut with **Sankalpa** and Hindi debut with **Ankur** . His notable works include **Malgudi Days** and several acclaimed Kannada films.
- **Recognition:** He has received **five Filmfare Awards** , **five Karnataka State Film Awards** , Karnataka Rajyotsava Award and Karnataka Sahitya Academy Award. He was honoured with the **Padma Bhushan** in 2025 .
- **Contributions:** He also served as a **Member of Karnataka Legislative Council (1988-1994)** and **Member of Legislative Assembly (1994-1999)** . He also served as **Minister of State for Bangalore City Development** .



## Dadasaheb Phalke Award

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- Instituted in **1969** , the award is India's **highest honour in the field of cinema** .
- It was introduced to commemorate **Dadasaheb Phalke** , the director of India's first full-length feature film **Raja Harishchandra (1913)** .
- The award recognises **outstanding contribution to the growth and development of Indian cinema** and includes a **Swarna Kamal (Golden Lotus) medallion, shawl and ₹10 lakh cash prize** .

Read More: [Dadasaheb Phalke Award](#)